PROJECT OVERVIEW STATEMENT	**Project Name:** Multimodal Misinformation Detector: Text and Image Fake-News Classification	**Student Name(s):** Ananya Saravani Ganni, Harika Etikela, Rafiul Haider

Problem/Opportunity:

Misinformation online increasingly spreads through two intertwined channels: fabricated or misleading text-based news content, and images used alongside legitimate or fabricated text to falsely lend credibility to a story. Automated fake-news text classifiers exist, but it is not well understood whether they are actually learning to detect factual inaccuracy or are instead relying on superficial stylistic patterns that merely correlate with misinformation. Within the image modality, prior work (e.g., Neves et al., 2020) has shown that AI-generated-image detectors trained on one generator's output can fail badly on a different, unseen generator, raising the question of whether combination strategies such as model ensembling and feature fusion genuinely improve generalization, or simply fit the training generator more precisely. Separately, it remains unclear whether combining textual and visual signals meaningfully improves detection over either modality alone, since a real image and a real caption can still be paired together in a misleading, out-of-context way that neither a text-only nor an image-only model can catch by itself.

This is our research question: How does a fine-tuned transformer's fake-news text classification accuracy compare to a classical TF-IDF baseline, and what do interpretability techniques (LIME/SHAP) reveal about which features each model relies on? Within the image modality, which combination strategy a single CNN, a reconstruction-error-based method, a multi-backbone CNN ensemble, or a CLIP-embedding feature-fusion classifier best classifies real versus AI-generated images, and does the best-performing in-distribution strategy maintain its advantage when tested on a genuinely unseen generator? Finally, how effectively can a fine-tuned multimodal model (Laya) detect out-of-context, misleading text-image pairings compared to a text-only version of the same model and a direct image-text embedding classifier?

Goal:

Build and deliver, within one semester, three complementary classification pipelines a text-based fake-news detector, an image-based AI-generated-image detector, and a combined text-image out-of-context detector each benchmarked against an appropriate baseline and paired with an interpretability analysis, converging on a shared evaluation of whether combining modalities (and, within the image track, combining models) improves misinformation detection over any single approach alone.

Objectives:

- **Objective 1 — Text Track: Cross-Dataset Generalization and Ensemble Reliability :** Train a TF-IDF+SVM baseline and a fine-tuned DistilBERT on WELFake, compare them in-domain, build an equal-weight soft-voting ensemble of their outputs, and apply LIME/SHAP to both models to compare feature reliance; completed by Submission 1 (baselines) and Submission 2 (ensemble, LIME/SHAP); measured by accuracy/precision/recall/F1 for all models in one comparison table plus a LIME/SHAP feature comparison; achieved using scikit-learn, Hugging Face transformers, LIME, and SHAP.
- **Objective 2 — Text Track: Cross-Dataset Generalization, Agreement, and Calibration:** Evaluate the Objective 1 models and ensemble, zero-shot, on ISOT and LIAR (LIAR's six labels mapped binary: pants-fire/false/barely-true = fake, half-true/mostly-true/true = real) to measure degradation from WELFake, whether SVM-DistilBERT agreement predicts correctness beyond individual confidence, and calibration (ECE, Brier); completed by Submission 2; measured by accuracy/precision/recall/F1 across all models and datasets in one table plus a written agreement/calibration analysis; achieved using scikit-learn, netcal, and Objective 1's model outputs.
- **Objective 3 — Image Track: Baseline, Combination Strategies, and Interpretability:** Train a CNN baseline (ResNet18 transfer learning), a reconstruction-error detector (autoencoder trained on real images only, following AEROBLADE), and two combination strategies a multi-backbone ensemble (ResNet18 + EfficientNet-B0) and a feature-fusion classifier (CLIP embeddings + autoencoder error, trained end-to-end) — to classify real vs. AI-generated images on CIFAKE, then apply Grad-CAM to the CNN baseline to visualize decision regions; completed by Submission 1 (baselines) and Submission 2 (combinations + Grad-CAM), measured by

accuracy/precision/recall/F1/AUC across all approaches in one comparison table plus at least 10 annotated Grad-CAM examples; achieved using PyTorch/torchvision, pytorch-grad-cam, and Hugging Face transformers (CLIP).

- **Objective 4 — Image Track: Generalization Test :** Evaluate every image-track approach, unchanged, on a small out-of-distribution sample of AI-generated images from a different generator family (StyleGAN-generated faces) than CIFAKE's Stable-Diffusion-based training data, completed by Submission 2, measured by reporting each approach's accuracy drop from in-distribution to out-of-distribution and identifying whether the best in-distribution combination strategy also generalizes best; achieved using the "140k Real and Fake Faces" dataset, directly testing whether ensembling/fusion strategies reduce generator-specific overfitting (per Neves et al., 2020) or merely improve in-distribution fit.
- **Objective 5 — Combined Track: Out-of-Context Detection with Fine-Tuned Laya:** Build a captioning pipeline that generates an image description without access to the article's caption, combine both texts into one record, and fine-tune two Laya variants (Attention and Gate) to predict consistent versus out-of-context, benchmarked across six comparison conditions against a text-only Laya baseline and a direct image-text embedding classifier on the NewsCLIPpings, VERITE, and MMFakeBench datasets, completed by Submission 2, measured by held-out test accuracy against stretch targets of 84.8% on NewsCLIPpings and 76.3% on VERITE (LAMAR-Attention and LAMAR-Gate targets per Papadopoulos et al., 2026 revision, arXiv:2504.06010v3) and a macro-F1 improvement over the text-only and embedding-classifier baselines, supported by paired 95% confidence intervals across three training seeds; achieved using a fine-tuned Laya model and scikit-learn, within a project compute.
- **Objective 6— Integrated Deliverable:** Deliver a working Streamlit application allowing a user to submit text, an image, or both, and receive a real/fake verdict with an interpretability explanation for each modality, plus a final report and poster synthesizing findings across all three tracks, completed by the Final Poster deadline, measured by a successful live demonstration and a coherent, integrated written narrative; achieved through coordinated documentation and a shared prediction-output schema across all three team members.

Success Criteria:

- Each track (text, image, combined) reports a clear, well-supported comparison between its baseline and alternative approaches, with results reproducible from the submitted code.
- The image track's generalization test provides a genuine, defensible answer to whether combination strategies (ensembling, feature fusion) improve robustness to an unseen generator, not just in-distribution accuracy.
- The interpretability analyses (LIME/SHAP for text, Grad-CAM for image) produce real, specific insight into what each model has learned — not just accuracy numbers.
- The combined track's out-of-context detection results report accuracy and macro-F1 with paired 95% confidence intervals across NewsCLIPpings, VERITE, and MMFakeBench, state plainly whether the accuracy stretch targets (84.8% NewsCLIPpings, 76.3% VERITE) were met, and clearly show whether fine-tuned Laya improves over the text-only and embedding-classifier baselines.
- The final Streamlit application successfully accepts text, image, or combined input and returns a verdict with an explanation for each modality submitted.
- The project is completed on the CS 668 course timeline, with all three team members able to independently explain and defend their own track during the in-class presentation.
- The final poster and report clearly communicate the limitations of each model alongside its strengths, and honestly distinguish which techniques are established in prior literature versus which comparisons/tests are this project's own contribution.
- The text track's ensemble generalization testprovides an empirical test of whether cross-model agreement between SVM and DistilBERT is associated with prediction correctness under distribution shift, beyond either model's individual confidence.

Assumptions, Risks, Obstacles:

- **Assumption:** The WELFake, CIFAKE, NewsCLIPpings, VERITE, MMFakeBench, and "140k Real and Fake Faces" datasets are large and clean enough to train and evaluate all three tracks' models without requiring additional data collection.
- **Assumption:** Free-tier compute (Google Colab GPU) is sufficient to fine-tune a transformer, train multiple CNN/autoencoder/fusion models, and compute CLIP embeddings within the semester timeline.
- **Risk:** Detectors trained on one AI image generator (e.g., Stable Diffusion, as used in CIFAKE) may not generalize to images from a different generator, a known failure mode documented in prior literature (Neves et al., 2020). Mitigation: this risk is directly addressed by generalization test rather than left unexamined.
- **Risk:** WELFake contains duplicate articles and may contain dataset- or source-specific stylistic artifacts that allow models to achieve high in-domain performance without necessarily demonstrating robust cross-dataset generalization, which may limit the robustness of in-domain evaluation. Mitigation: zero-shot evaluation on LIAR/ISOT directly tests whether performance survives outside WELFake's specific data distribution.
- **Obstacle:** The three tracks are being developed in parallel by three different team members; the combined track depends on outputs from both the text and image tracks being finalized first. Mitigation: agree on a shared model-output schema (label, confidence score) early, and prioritize finishing single-modality baselines before the combined track's development begins.
- **Obstacle:** The combined track depends on a working image-captioning pipeline, reliable access to the NewsCLIPpings, VERITE, and MMFakeBench images/captions, and free-tier GPU availability; missing images, overlap between the three datasets, incomplete or invented image descriptions, and unreliable free compute could each delay the Submission 2 evaluation. Mitigation: confirm dataset access early, audit for identifier/hash overlap across datasets, fix and cache one captioning checkpoint's outputs, pilot all six comparison conditions on a small sample before committing to full training, and reserve paid compute as a fallback.
- **Obstacle:** NewsCLIPpings and VERITE are harder, more realistic cross-modal tasks than standard fake-news classification, and even the stretch targets (84.8% NewsCLIPpings, 76.3% VERITE) represent published state-of-the-art results rather than guaranteed outcomes; the team should not expect CIFAKE- or WELFake-level accuracy on this track. Mitigation: report unmet performance targets transparently rather than treating them as a failure, and prioritize the core NewsCLIPpings comparison over the additional benchmarks if time runs short.

Prepared By	**Date**	**Approved By**	**Date**
Ananya Saravani Ganni, Harika Etikela, Rafiul Haider	09-28-2026		